

Fog Detection: The Optical Route



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A majority of serial collision accidents happens when visibility is low due to fog. The risk of these accidents can be reduced considerably with a system containing visibility sensors and variable speed-limit signs.

Until now, such systems have been very expensive because of the high cost of visibility sensors. Fortunately, these days, availability of reasonably-priced optical fog sensors (OFSes) changes the calculus for such systems in a drastic manner.

How does an OFS work

Backscattering is the principle behind OFSes. It is the reflection of waves, particles or signals back to the direction from which these came. It is diffuse reflection due to scattering, as opposed to specular reflection like in a mirror.

In a typical OFS, a narrow beam of a red laser light (eye-safe laser) comes out of an opening on the front. A detector behind a lens in another window is sensitive to incoming laser light in a narrow lobe that overlaps the transmitter beam.

If there are fog particles in the overlap zone, light is scattered back and a signal is induced on the sensor's raw signal output.

The raw signal output is an analogue voltage and a measure of the amount of backscattered light from the overlap (sensitive) zone. So the more fog in the overlap zone, the stronger the signal.

Density of fog, which is the function of visibility, can be measured within a controlled environment using an OFS as outlined above. Here,

signal analysis/processing is done in an integrated microcontroller (MCU). The primary signal is first sampled for a finite duration and then calculated according to a derived formula, to convert measured signal into visibility.

Calculated visibility is presented in digital form (as an ASCII string) on the serial (RS232) output of the OFS. The MCU also controls the analogue output, indicating visibility directly. This analogue output, like digital output, is also updated at regular intervals.

In practice, light sender and receiver optics are heated up to (slightly above) the ambient-temperature level using a heating wire (anti-dew heater) mounted around these. This is done to ensure that no water droplets form, thus no further (undesired) losses are incurred due to absorption and refraction of the transmitted light beam.

Besides, in order to keep electronics dry, a membrane ventilator is used that keeps the pressure inside at the same level as outside. This prevents liquid (water) from getting sucked into the sensor through microcracks at a falling temperature.

Applications

It is well-known that fog causes many accidents. An OFS will work well for road safety and a lot of other applications. The sensor is sensitive to particles in a limited zone ahead of the location of the sensor that limits the visibility in the air.

These particles are normally microscopic water particles constituting fog, but these could also be snowflakes, raindrops or air pollutants. However, main indications of low visibility are fog or snow.

Exterior scene. Road keepers often install OFSes because they need information about the weather-associated



Fig. 1: Optical fog sensor

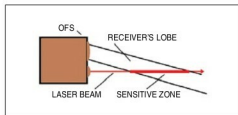


Fig. 2: Backscattering technique in an OFS

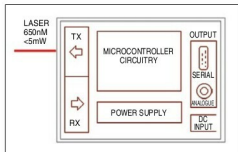


Fig. 3: Typical system diagram of an OFS